

CO1 AT2 Application-Based Questions

Course Code: DSA0502

Course Name: Query processing for data science

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Que 37. Understanding a Database Schema

A university database contains the following tables:

- ☐ Student(StudentID, Name, DeptID)
- ☐ Department(DeptID, DepartmentName)

Task:

- ☐ Identify the primary key and foreign key.
- ☐ Explain the relationship between the two tables.
- ☐ Merge the tables using Python to display each student's department name.

StudentID Name DeptID

1	Arun	101
2	Bala	102
3	Chitra	101
4	Divya	103

DeptID DepartmentName

101	CSE
102	ECE
103	AI & DS

- ☐ Identify the primary key and foreign key.

Table	Primary Key	Foreign Key
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Table	Primary Key	Foreign Key
Student(StudentID, Name, DeptID)	StudentID	DeptID (references Department table)
Department(DeptID, DepartmentName)	DeptID	None

Explain the relationship between the two tables.

Department

DeptID DepartmentName	
101	CSE
102	ECE
103	AI & DS
One-to-Many	
V	

Student

StudentID Name DeptID		
1	Arun	101
2	Bala	102
3	Chitra	101
4	Divya	103

Merge the tables using Python to display each student's department name.

```
1 import pandas as pd
2 # Student table
3 student = pd.DataFrame({
4     "StudentID": [1, 2, 3, 4],
5     "Name": ["Arun", "Bala", "Chitra", "Divya"],
6     "DeptID": [101, 102, 101, 103]
7 })
8 # Department table
9 department = pd.DataFrame({
10     "DeptID": [101, 102, 103],
11     "DepartmentName": ["CSE", "ECE", "AI & DS"]
12 })
13 # Merge the tables
14 result = pd.merge(student, department, on="DeptID")
15
16 print(result)
```

Output:

	StudentID	Name	DeptID	DepartmentName
0	1	Arun	101	CSE
1	2	Bala	102	ECE
2	3	Chitra	101	CSE
3	4	Divya	103	AI & DS

Que:19. Smart City Traffic Data

Traffic information is stored in traffic.csv.

Task:

- ☐ Calculate average daily traffic.
- ☐ Find roads with the highest congestion.
- ☐ Display accident-prone locations.

Input Table (traffic.csv)

Date	Road	Daily_Traffic	Congestion_Level	Accident_Location
01-07-2026	Anna Salai	12000	High	Teynampet
01-07-2026	OMR	15000	Very High	Sholinganallur

Date	Road	Daily_Traffic	Congestion_Level	Accident_Location
01-07-2026	GST Road	10000	Medium	Guindy
02-07-2026	Anna Salai	12500	High	Teynampet
02-07-2026	OMR	16000	Very High	Navalur
02-07-2026	GST Road	9800	Medium	Guindy
03-07-2026	Anna Salai	13000	High	Saidapet
03-07-2026	OMR	17000	Very High	Sholinganallur
03-07-2026	GST Road	10500	Medium	Tambaram
04-07-2026	Anna Salai	12800	High	Teynampet

Code:

```
C: > Users > HP > Downloads > exp-7.py > ...
1 import pandas as pd
2
3 traffic = pd.read_csv(r"C:\Users\HP\Downloads\dataset at1cot2.txt", sep="\t")
4
5 print(traffic.columns)
6 # 1. Calculate average daily traffic
7 average_traffic = traffic["Daily_Traffic"].mean()
8 print("Average Daily Traffic:", average_traffic)
9
10 # 2. Find roads with the highest congestion
11 highest_congestion = traffic[traffic["Congestion_Level"] == "Very High"]
12 print("\nRoads with Highest Congestion:")
13 print(highest_congestion[["Road", "Congestion_Level"]])
14
15 # 3. Display accident-prone locations
16 accident_locations = traffic["Accident_Location"].value_counts()
17 print("\nAccident-Prone Locations:")
18 print(accident_locations)
```

Output:

```
Average Daily Traffic: 12130.0

Roads with Highest Congestion:
   Road Congestion_Level
1   OMR      Very High
5   OMR      Very High
9   OMR      Very High
13  OMR      Very High
17  OMR      Very High
```

```
python3 -c "import sys; sys.stdout.write('Accident-Prone Locations:'); sys.stdout.write('Accident_Location'); sys.stdout.write('Teynampet 3'); sys.stdout.write('Sholinganallur 3'); sys.stdout.write('Guindy 3'); sys.stdout.write('Neelankarai 2'); sys.stdout.write('Injambakkam 2'); sys.stdout.write('Saidapet 2'); sys.stdout.write('Tambaram 2'); sys.stdout.write('Navalur 1');"
```